

Causality - SoSe 2025

Score-Based Causal Discovery

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Last lecture, we discussed the following concepts

- Identification results for restricted models
 - General non-identifiability
 - LiNGAM, and extensions
 - RESIT
- Causal orderings and linear ICA

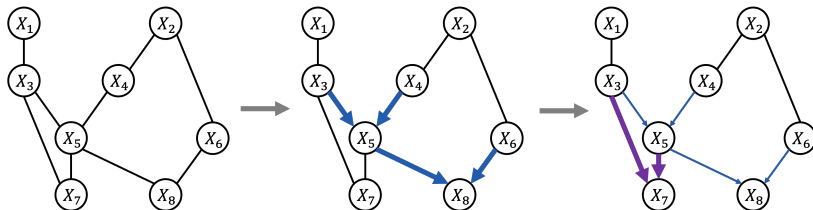
Questions:

- What are the key assumptions in constraint-based causal discovery?
 - Markov, Faithfulness, (perfect/good) independence test
- Which graph structures can the PC or the SGS algorithm identify?
 - CPDAG (Markov equivalence class)
- Which graph structures can the ICA-LiNGAM identify?
 - Causal order, DAG (after pruning)
- What are the key assumptions of LiNGAM?
 - Restricted SCM (linear, non-Gaussian), structure minimality, ICA algorithm, (perfect/good) independence test

So far (SGS/PC)

Algorithm (sketch):

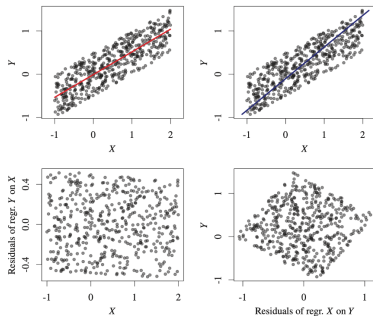
1. Skeleton search
2. Identify v-structures
3. Apply further orientation rules



So far (LiNGAM/RESIT)

Algorithm for Pairs (sketch):

1. Perform **regression** in both directions
2. Assess **independence** between residuals and potential cause
3. If there is only one direction which admits an independence, chose the corresponding graph. Otherwise, if a decision is forced, decide for direction with higher p -value (less evidence to reject independence)



Today we take a look at the last class of causal discovery methods, i.e., score-based methods:

- Score-based causal discovery
- Greedy Equivalence Search (GES)
- Searching efficiently through DAGs (CAM, NOTEARS)

Causal Discovery: Overview

Constraint-Based

- Measure **conditional independencies**
- Search for graph which **satisfies** them

Examples

SGS, PC

Identifies

CPDAG

(Markov equivalence class)

Score-Based

- **Score** models based on data fit
- Large search space, greedy approach

Examples

GES

Identifies

CPDAG

(Markov equivalence class)

Functional/Restricted

- **Additional assumptions** on model class, noise distribution, etc.

Examples

LiNGAM, RESIT, CAM

Identifies

DAG

(Usually: True model)

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Score-Based Causal Discovery

Score-Based Causal Discovery

Idea: Treat structure learning as a model selection problem.

- Given n iid observations $\mathcal{D} = (\mathbf{x}_V^{(1)}, \dots, \mathbf{x}_V^{(n)})$, assign a **score** $\mathcal{S}(\mathcal{D}, \mathcal{G})$ to each graph $\mathcal{G}$
- $\mathcal{S}(\mathcal{D}, \mathcal{G})$ measures how well $\mathcal{G}$ fits the data
- Search over the space of DAGs to find the graph that maximizes:

$$\hat{\mathcal{G}} = \operatorname{argmax}_{\mathcal{G}} \mathcal{S}(\mathcal{D}, \mathcal{G})$$

Score-Based Causal Discovery

How can we define a score $\mathcal{S}(\mathcal{D}, \mathcal{G})$?

- Assume a **parametric model**
 - E.g. linear Gaussian equations
 - Introduces a set of parameters θ
- Consider the **maximum likelihood estimator** (MLE) $\hat{\theta}$ for θ

Idea 1: Use **log-likelihood** $\log p(\mathcal{D}|\hat{\theta}, \mathcal{G})$ for the score
(the log-likelihood is factorized according to the given graph)

$$\log p(\mathcal{D}|\hat{\theta}, \mathcal{G}) = \sum_{i=1}^d \sum_{j=1}^n \log p(x_i^{(j)} | x_{\text{Pa}_{\mathcal{G}}(i)}^{(j)}, \hat{\theta})$$

Issue: Learns full DAG since more parameters always improve the model fit

Solution: use **penalized** log-likelihood

Score-Based Causal Discovery

How can we define a score $\mathcal{S}(\mathcal{D}, \mathcal{G})$?

- Assume a **parametric model**
 - E.g. linear Gaussian equations
 - Introduces a set of parameters θ
- Consider the **maximum likelihood estimator** (MLE) $\hat{\theta}$ for θ

Idea 2: Use the **Bayesian Information Criterion (BIC)**

$$\mathcal{S}(\mathcal{D}, \mathcal{G}) = \underbrace{2 \log p(\mathcal{D} | \hat{\theta}, \mathcal{G})}_{\text{log-likelihood}} - \underbrace{\log n \cdot (\# \text{ parameters})}_{\text{penalty}}$$

- The penalty induces sparsity in the DAG penalizes the number of parameters ($\propto$ edges)
- Other possibilities such as AIC exist

Properties of BIC

- **Score-equivalence:** Same score for all DAGs within the same MEC (linear-Gauss, or categorical)
- **Decomposable:** Score can be computed as a sum of individual terms for each node given its parents (capital X_i contains all observations)

$$\mathcal{S}(\mathcal{D}, \mathcal{G}) = \sum_{i=1}^d s(X_i, \mathbf{X}_{\text{Pa}(i)})$$

where s is the local score (log-likelihood) $\Rightarrow$ **Computational savings**

- **Local consistency:** Score of a DAG model $\mathcal{G} \dots$
 - **increases** when adding an edge that eliminates an independence constraint that does not hold in P
 - **decreases** when adding an edge that does not eliminate such a constraint (i.e., adding to an ancestor j of i , and $j \notin \text{Pa}(i)$)

Example

An example and a derivation for BIC of a linear Gaussian SCM are provided in the lecture notes.

Scoring DAGs

So now we can just score all DAGs and learn the CPDAG?

#Nodes	#Possible DAGs
0	1
1	1
2	3
3	25
4	543
5	29281
6	3781503
7	1138779265
8	783702329343
9	1213442454842881
10	4175098976430598143
11	31603459396418917607425
12	521939651343829405020504063
13	18676600744432035186664816926721
14	1439428141044398334941790719839535103

Figure 1: Figure from: <https://oeis.org/A003024> (February 2022)

Greedy Equivalence Search

Greedy Equivalence Search

- Optimizes BIC
- Searches over **Markov equivalence classes** instead of DAGs
- Assumes Markov and faithfulness

Greedy Equivalence Search (GES)

- MECs are represented by the corresponding CPDAG
- Start with *empty* class (empty CPDAG)
- Two phases:
 1. **Add edges** until local maximum is reached
 2. **Remove edges** until local maximum is reached
- Output the CPDAG attaining the maximum in Phase 2

GES Neighborhood Relation

Denote $\mathcal{E}$ as the Markov equivalence class of a DAG model.

Neighborhood relation in **Phase 1** of GES:

- $\mathcal{E}^+(\mathcal{E})$: neighbors of state $\mathcal{E}$ in Phase 1
- An equivalence class $\mathcal{E}'$ is in $\mathcal{E}^+(\mathcal{E}) \Leftrightarrow$ there is some DAG $\mathcal{G} \in \mathcal{E}$ to which we can **add** a single edge that results in a DAG $\mathcal{G}' \in \mathcal{E}'$

Neighborhood relation in **Phase 2** of GES:

- $\mathcal{E}^-(\mathcal{E})$: neighbors of state $\mathcal{E}$ in Phase 2
- An equivalence class $\mathcal{E}'$ is in $\mathcal{E}^-(\mathcal{E}) \Leftrightarrow$ there is some DAG $\mathcal{G} \in \mathcal{E}$ from which we can **remove** a single edge that results in a DAG $\mathcal{G}' \in \mathcal{E}'$

Phase 1 Neighborhood Relation - Example

An equivalence class $\mathcal{E}'$ is in $\mathcal{E}^+(\mathcal{E})$
 $\Leftrightarrow$ there is some DAG $\mathcal{G} \in \mathcal{E}$ to
which we can **add** a single edge that
results in a DAG $\mathcal{G}' \in \mathcal{E}'$

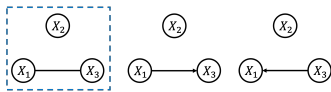


Figure 2: Current class $\mathcal{E}$

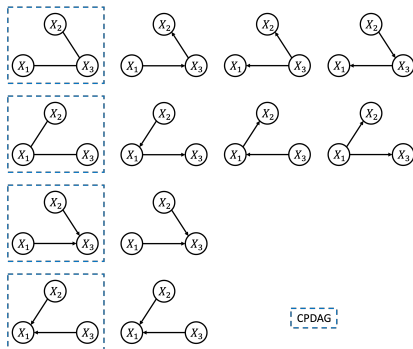


Figure 3: Neighboring classes $\mathcal{E}^+(\mathcal{E})$

- Scoring: score a representative DAGs of $\mathcal{E}$ (**score equivalence**)
- Compute change in score locally: Only one node changes its parents in each step (**decomposability**)

Phase 2 Neighborhood Relation - Example

Take any DAG from $\mathcal{E}^+(\mathcal{E})$, e.g., $\mathcal{G}' \in \mathcal{E}'$ referring to $X_2 \rightarrow X_1 \rightarrow X_3$.

Which of the following statements are true?

1. $\mathcal{E}^-(\mathcal{E}')$ is again equal to $\mathcal{E}$.
2. $\mathcal{E}^-(\mathcal{E}')$ contains two equivalence classes with two members each.
3. $\mathcal{E}^-(\mathcal{E}')$ contains three equivalence classes with two members each.

Answer:

- Only (2.) is true. Two possible CPDAGs with two members each.

GES Example

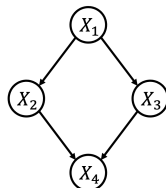
In the following, we show an example run¹ of GES with BIC for linear Gaussian models applied to an iid sample of size 1 000 drawn from the SCM (all N_i are standard Gaussian and independent of each other):

$$X_1 := N_1$$

$$X_2 := 0.5 \cdot X_1 + N_2$$

$$X_3 := 0.5 \cdot X_1 + N_3$$

$$X_4 := 1.5 \cdot X_2 + 0.7 \cdot X_3 + N_4$$

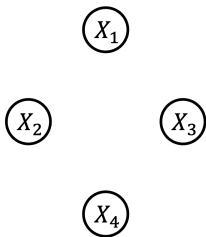


Question: Do you expect that we need Phase 2 for this DAG? If yes, why?

¹We attach the Python code “lecture_10_code.py” to reproduce the results (which unfortunately does not keep track of the score for each step).

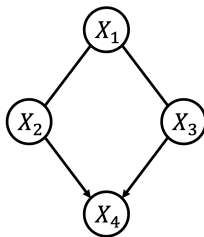
GES Example Run

Current estimate



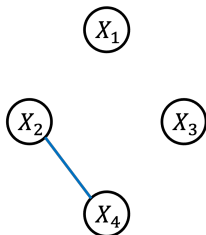
(Phase I—forward)

True equivalence class



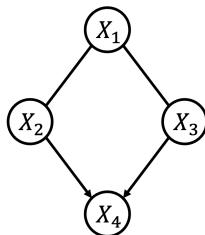
GES Example Run

Current estimate



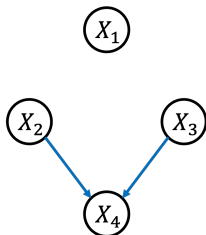
(Phase I—forward)

True equivalence class



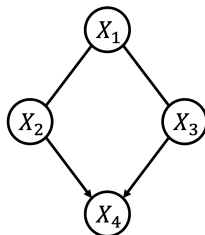
GES Example Run

Current estimate



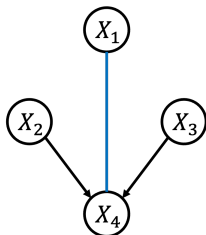
(Phase I—forward)

True equivalence class



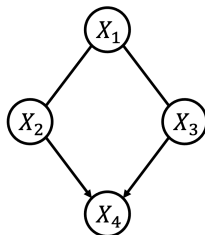
GES Example Run

Current estimate



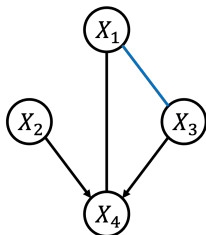
(Phase I—forward)

True equivalence class



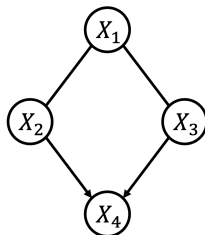
GES Example Run

Current estimate



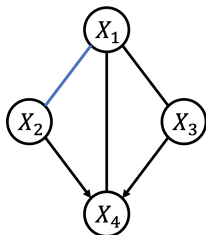
(Phase I—forward)

True equivalence class



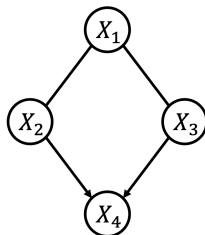
GES Example Run

Current estimate



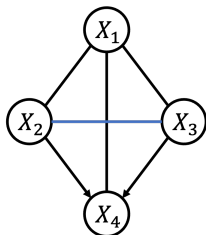
(Phase I—forward)

True equivalence class



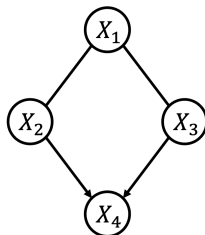
GES Example Run

Current estimate



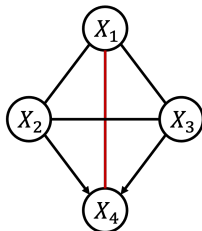
(Phase I—forward)

True equivalence class



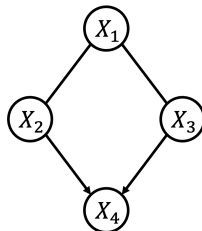
GES Example Run

Current estimate



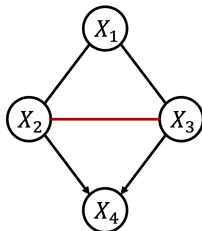
(Phase II—backward)

True equivalence class



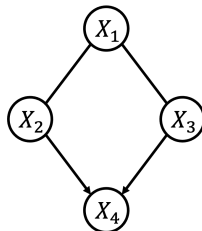
GES Example Run

Current estimate



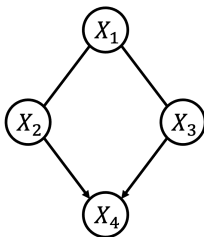
(Phase II—backward)

True equivalence class



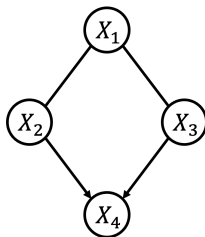
GES Example Run

Current estimate



(Final estimate)

True equivalence class



Greedy Equivalence Search (GES)

- MECs are represented by the corresponding CPDAG
- Start with *empty* class (empty CPDAG)
- Two phases:
 1. **Add edges** until local maximum is reached
 2. **Remove edges** until local maximum is reached
- Output the CPDAG attaining the maximum in Phase 2

Phase 1: The hope is to identify a model that is as simple as possible

- Ignoring computational costs, one could also start with a full DAG and only do Phase 2
- Chickering (2002): GES identifies the true CPDAG as $n \rightarrow \infty$
 - Requires appropriate scoring criterion (e.g., BIC for categorical or linear Gaussian data)

Scoring DAGs

Question: Can we learn DAGs based on scores?

- For models for which we can specify the full likelihood, which are **identifiable**, yes!
 - For example, non-linear Gaussian additive noise models

Causal Additive Models (CAM)

Definition (Bühlmann et al. 2014) p_X admits an **causal additive model (CAM)**, if for all $j \in \mathbf{V}$

$$X_i := \sum_{k \in \text{Pa}(i)} f_{i,k}(X_k) + N_i ,$$

where all N_i, N_j for $i \neq j$ are pairwise independent and Gaussian distributed (specific type of ANM).

- We know that for **non-linear** $f_{i,k}$, the model is identifiable
- We can derive the likelihood of the model

Algorithm Sketch:

1. Determine causal order that maximizes the likelihood
2. Apply pruning, e.g., based on significance testing to obtain a DAG

NOTEARS (Zheng et al. 2018)

Since searching over DAGs is infeasible, Zheng et al. (2018) derive a continuous DAG constraint.

Problem: In score-based DAG learning, we need to solve the following optimization problem

$$\hat{\mathcal{G}} = \operatorname{argmax}_{\mathcal{G}} \mathcal{S}(\mathcal{D}, \mathcal{G}) \quad \text{subject to} \quad \underbrace{\mathcal{G} \text{ is a DAG}}_{\text{combinatorial constraint}}$$

which is infeasible for larger graphs.

NOTEARS (Zheng et al. 2018)

Since searching over DAGs is infeasible, Zheng et al. (2018) derive a continuous DAG constraint.

Key Idea: They derive a continuous constraint based on a learned weighted adjacency matrix $\mathbf{W}$, which they show is zero if and only if $\mathbf{W}$ encodes a DAG, i.e.,

$$\hat{\mathbf{W}} = \operatorname{argmax}_{\mathbf{W}} \mathcal{S}(\mathcal{D}, \mathbf{W}) \quad \text{subject to} \quad \underbrace{\operatorname{tr}(e^{\mathbf{W} \circ \mathbf{W}})}_{\text{continuous constraint}} - d = 0$$

where the score can be computed based on $\mathbf{W}$.

- If our score encodes an **identifiable model** (like for CAM), we can simply use gradient descent with the above objective to learn a DAG!

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(Usually: True model)

- **Recommended reading** ECI book Chapter 7.2.
- This course *builds upon and extends* the material from “Causality” (WiSe 2022/23) that was kindly provided by **Stefan Harmeling**
- This lecture *builds upon and extends* the slides of the course “Causality” (2021) by Christina Heinze-Deml
- Chickering. *Optimal Structure Identification With Greedy Search*. JMLR, 2002
- Bühlmann et al. *CAM: Causal additive models, high-dimensional order search and penalized regression*. Annals of Statistics, 2014
- Zheng et al. *DAGs with NO TEARS: Continuous Optimization for Structure Learning*, NeurIPS, 2018

Thank you!